

1. A 7-year-old girl is brought to the physician because of progressive sleepiness, ataxia, and double vision over the past 2 days. She has a 2-year history of complex partial seizures treated with carbamazepine. One week ago, she developed fever and received erythromycin for pharyngitis. Her brother has attention-deficit/hyperactivity disorder treated with methylphenidate. Her temperature is 37°C (98.6°F). She appears sleepy. Ocular movements appear full. She has no weakness, but is moderately ataxic. Which of the following is the most likely cause of her current symptoms?
- ☐ A) Carbamazepine-erythromycin interaction
 - ☐ B) Cerebellar astrocytoma
 - ☐ C) Erythromycin allergy
 - ☐ D) Methylphenidate ingestion
 - ☐ E) Viral encephalitis



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

2. A 52-year-old man comes to the physician because of progressive bilateral leg pain over the past year. The pain is located on the posterior aspect of the thighs and calves, occurs only when he is standing, and resolves within 30 seconds of sitting or lying down. He has noticed that the pain is less intense if he leans on something as he walks, such as a grocery cart. He can still ride an exercise bicycle without developing any pain. He has hypertension, hyperlipidemia, and a small abdominal aortic aneurysm. He had a left carotid endarterectomy 3 years ago after several transient ischemic attacks in the left hemisphere. Medications include lisinopril, lovastatin, and aspirin. He has smoked one pack of cigarettes daily for 30 years. He is 180 cm (5 ft 11 in) tall and weighs 110 kg (242 lb); BMI is 34 kg/m². His temperature is 37°C (98.6°F), pulse is 70/min, respirations are 12/min, and blood pressure is 150/90 mm Hg. Examination shows a right femoral bruit and diminished pedal pulses. Which of the following is the most likely cause of this patient's lower extremity pain?

- ☐ A) Cerebrovascular disease
- ☐ B) Lumbar spinal stenosis
- ☐ C) Lumbosacral plexopathy
- ☐ D) Peripheral vascular disease
- ☐ E) Trochanteric bursitis



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

3. A 57-year-old woman comes to the physician because of a 2-day history of dribbling fluid from her mouth every time she drinks. She has not had difficulty swallowing or any other symptoms. She has no history of serious illness and takes no medications. She has an active lifestyle. Vital signs are within normal limits. Examination shows drooping of the lower lip on the left. The left forehead does not wrinkle on upward gaze. The remainder of the examination shows no abnormalities. Which of the following is the most appropriate next step in management?

- ☐ A) Echocardiography
- ☐ B) MRI of the brain
- ☐ C) Doxycycline therapy
- ☐ D) Prednisone therapy
- ☐ E) Lumbar puncture



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

4. A 72-year-old man comes to the physician because of a 1-week history of imbalance and leg weakness and a 3-day history of difficulty initiating urination. He also has had mild midthoracic back pain for 5 months, but during the past 2 weeks, the pain has worsened and frequently awakens him. He has had a 9-kg (20-lb) weight loss since an episode of bronchitis 6 months ago. He has smoked one pack of cigarettes daily for 50 years. He is cachectic and appears chronically ill. Examination shows scleral icterus. Muscle strength in the lower extremities is decreased in the proximal and distal muscles bilaterally. There is no muscle atrophy, but tone is increased. Deep tendon reflexes are hyperactive, and Babinski sign is present bilaterally. Sensation is decreased to pinprick to the umbilicus. Muscle strength, sensation, and reflexes in the upper extremities are normal. Which of the following is the most likely diagnosis?

- ☐ A) Epidural spinal cord compression
- ☐ B) Myasthenic (Lambert-Eaton) syndrome
- ☐ C) Spinal cord infarction
- ☐ D) Syringomyelia
- ☐ E) Transverse myelitis



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

5. A 67-year-old woman comes to the physician because of a 6-month history of buttock and leg pain. The pain occurs when she walks for more than two blocks or stands for more than 10 minutes; sitting or bending forward provides relief within 1 minute. She has not had weakness or loss of sensation. Bladder function has been normal. She has a 20-year history of lower back pain relieved with rest. She has type 2 diabetes mellitus well controlled with metformin. Muscle strength is full. Deep tendon reflexes are 2+ except for absent ankle reflexes. Sensation is intact. There is decreased lumbar lordosis; lumbar flexion is full and extension is limited. Straight-leg raising is negative. Peripheral pulses are intact. Which of the following is the most likely diagnosis?

- ☐ A) Femoropopliteal arterial stenosis
- ☐ B) Lumbar spinal stenosis
- ☐ C) Piriformis syndrome
- ☐ D) Sacroiliac arthritis
- ☐ E) Trochanteric bursitis



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

6. An otherwise healthy 77-year-old woman comes to the physician because of a 1-month history of severe dizziness. Several times every day she has the sudden onset of a spinning sensation and loss of equilibrium that lasts for 20 to 30 seconds. The episodes begin most frequently after she rolls over in bed or turns her head to the right. Her pulse is 76/min, and blood pressure is 140/80 mm Hg. Neurologic examination shows no abnormalities. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) Positional oculovestibular testing
- ☐ B) Tilt table test
- ☐ C) Holter monitoring
- ☐ D) Doppler ultrasonography of the carotid arteries
- ☐ E) EEG
- ☐ F) Electronystagmography
- ☐ G) Magnetic resonance angiography



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

7. A 35-year-old woman has had constant burning pain in her right hand since closing her hand in a car door 1 year ago. Examination immediately following the injury showed extensive ecchymosis and swelling. No fractures were identified, but the hand was placed in a cast for 1 month. She is unable to tolerate light touch or any motion of the wrist or finger joints during the current examination. Which of the following is the most likely explanation for these findings?

- ☐ A) Abnormal sympathetic neural response
- ☐ B) Increased intracarpal canal pressure
- ☐ C) Median nerve fibrosis
- ☐ D) Post-traumatic arthritis
- ☐ E) Radial artery thrombosis



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

8. A 25-year-old woman has had generalized weakness for 2 days. On questioning, she reports numerous hospital admissions for abdominal pain. One episode occurred after treatment of a urinary tract infection with a sulfonamide. Examination shows global weakness and areflexia; Babinski sign is absent. Sensation to pinprick is mildly decreased in a stocking-glove distribution, and sensation to vibration is decreased over the feet. She has tachycardia that does not slow with carotid massage. Which of the following is the most likely diagnosis?

- ☐ A) Acute intermittent porphyria
- ☐ B) Alcoholic neuropathy
- ☐ C) Charcot-Marie-Tooth disease
- ☐ D) Diabetic radiculopathy
- ☐ E) Guillain-Barré syndrome
- ☐ F) Leprosy
- ☐ G) Paraneoplastic neuropathy



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

9. A 62-year-old woman comes to the physician because of a 6-week history of stiffness and pain of her neck, shoulders, and hips. She has difficulty dressing herself and combing her hair in the morning. She also has had generalized weakness, malaise, and a low-grade fever. She has had a 2.3-kg (5-lb) weight loss during the past month despite no changes in diet. She has no history of serious illness. Her only medication is occasional acetaminophen for pain. Vital signs are within normal limits. Examination shows pitting edema of the hands and the dorsal aspects of the feet. Range of motion of the neck, shoulders, and hips is limited. Her erythrocyte sedimentation rate is 110 mm/h. Additional laboratory findings, including her serum thyroid-stimulating hormone concentration, are within the reference ranges. X-rays of the shoulders and feet show no abnormalities. Which of the following is the most likely diagnosis?

- ☐ A) Bursitis
- ☐ B) Dermatomyositis
- ☐ C) Fibromyalgia
- ☐ D) Polymyalgia rheumatica
- ☐ E) Polymyositis
- ☐ F) Rheumatoid arthritis



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

10. A 57-year-old woman comes to the physician because of daily headaches for 8 months. The headaches are localized to the right temple and usually worsen throughout the day. At times, the headaches are exacerbated by chewing. She has a history of migraines but has not had one since menopause 6 years ago. She recently divorced and is under a great deal of stress at work. She has smoked one-half pack of cigarettes daily for 25 years. Her temperature is 37°C (98.6°F), pulse is 82/min, and blood pressure is 144/86 mm Hg. Examination shows pain with movement of the jaw and tenderness under the right zygomatic arch. The nasal mucosa is normal. Erythrocyte sedimentation rate is 20 mm/h. Which of the following is the most likely diagnosis?

- ☐ A) Bacterial meningitis
- ☐ B) Cluster headache
- ☐ C) Migraine
- ☐ D) Sinusitis
- ☐ E) Temporal arteritis
- ☐ F) Temporomandibular joint syndrome
- ☐ G) Tension-type headache
- ☐ H) Trigeminal neuralgia



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

11. A 42-year-old man comes to the physician with his wife because of a 6-month history of increasing clumsiness and a 1-month history of irritability. His wife has noticed jerkiness of his limbs when he walks down stairs or walks quickly. During the past month, he has had the same involuntary movements at rest after stressful days at work. He has not had fever, chills, or night sweats. He does not smoke cigarettes, drink alcohol, or use illicit drugs. He goes camping several times yearly. He does not appear to be in distress. Examination shows involuntary movements of the right upper extremity, which the patient is able to suppress when made aware of them. His gait is ataxic. Additional testing is most likely to show which of the following abnormalities?

- ☐ A) Atrophy of the caudate nucleus on brain imaging
- ☐ B) Decreased serum methylmalonic acid concentration
- ☐ C) Dilated ventricles on brain imaging
- ☐ D) Increased oligoclonal bands in cerebrospinal fluid
- ☐ E) Increased serum acetylcholine esterase inhibitor concentration
- ☐ F) Positive serum IgM Lyme titer



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

12. A previously healthy 82-year-old man is brought to the emergency department because of a 3-hour history of difficulty communicating. His wife reports that when he awoke this morning, he seemed slightly confused and "did not make sense." He has hyperlipidemia treated with simvastatin. He does not appear to be in distress. He is 170 cm (5 ft 7 in) tall and weighs 100 kg (220 lb); BMI is 35 kg/m². His temperature is 37°C (98.6°F), pulse is 85/min and irregular, respirations are 15/min, and blood pressure is 145/90 mm Hg. Examination shows mild right pronator drift. During the examination, he is attentive but has some difficulty naming objects correctly and following two-step commands. An ECG shows atrial fibrillation. Which of the following is this patient's greatest risk factor for his neurologic condition?

- ☐ A) Age
- ☐ B) Dysrhythmia
- ☐ C) Hyperlipidemia
- ☐ D) Increased blood pressure
- ☐ E) Weight



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

13. A 77-year-old man is brought to the physician by his daughter because he has become increasingly forgetful during the past 6 months. His daughter states that his short-term memory is quite impaired, and he says that he is frustrated with his memory. He lives alone and is able to perform activities of daily living independently. On questioning, he reports a 1-month history of a mild nonproductive cough, constipation, mild pain in his hips with movement, mild numbness of the toes of both feet, and a few episodes of urinary incontinence. He has hypertension, hyperlipidemia, and well-controlled type 2 diabetes mellitus. Current medications include metformin, lisinopril, atorvastatin, and aspirin. His wife died 6 years ago. He does not smoke or drink alcohol. His temperature is 36.8°C (98.2°F), pulse is 72/min, respirations are 14/min, and blood pressure is 128/72 mm Hg. Muscle strength is normal. Cranial nerves are intact. Deep tendon reflexes are normal. Sensation to vibration is mildly decreased over the distal lower extremities. Proprioception is intact. His gait is wide based, and he has impaired tandem walking. His Mini-Mental State Examination score is 24/30; he recalls one of three objects after 5 minutes. An MRI of the brain shows slightly enlarged ventricles with minimal cortical atrophy. Which of the following is the most appropriate next step in management?
- ☐ A) Recommendation that the patient move to an assisted living community
 - ☐ B) Discontinuation of atorvastatin therapy
 - ☐ C) Discontinuation of metformin therapy
 - ☐ D) Addition of donepezil therapy
 - ☐ E) Addition of paroxetine therapy
 - ☐ F) Lumbar puncture



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

14. A previously healthy 42-year-old man comes to the physician because of a 6-week history of intermittent pain and tingling that radiates from his right shoulder blade to his right thumb and index finger. These symptoms worsen with coughing and sneezing. There is no history of neck, shoulder, or arm injury. His pulse is 72/min, and blood pressure is 120/70 mm Hg. Physical examination shows no abnormalities. The cranial nerves are intact. Muscle strength is normal in all extremities. Deep tendon reflexes are 1+ in the right biceps muscle and 2+ elsewhere. Babinski sign is absent. Sensation to pinprick is decreased over the dorsal aspect of the right thumb and index finger. Which of the following is the most likely mechanism of this patient's symptoms?

- | | |
|--|--|
| ☐ A) Brachial plexus compression | ☐ G) Median nerve compression at the wrist |
| ☐ B) Central nervous system compression | ☐ H) Radial nerve compression at the elbow |
| ☐ C) Central nervous system demyelination | ☐ I) Radial nerve compression at the spiral groove |
| ☐ D) Central nervous system ischemia | ☐ J) Ulnar nerve compression at the elbow |
| ☐ E) Cervical nerve root compression | ☐ K) Ulnar nerve compression at the wrist |
| ☐ F) Median nerve compression at the elbow | |



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

15. A 67-year-old man with hypertension is brought to the emergency department 30 minutes after a stroke. Examination shows Horner syndrome and paralysis of the palate on the right. Sensation is decreased to pain over the right side of the face and left side of the body, and sensation is absent to hot and cold over the right face and left side of the body. There is no paralysis of the extremities. Which of the following is the most likely site of infarction?

- ☐ A) Right brachium pontis
- ☐ B) Right lateral medulla
- ☐ C) Right midbrain
- ☐ D) Right temporal lobe
- ☐ E) Right thalamus



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

16. A 5-year-old girl is brought to the emergency department by her parents because of a 4-hour history of unsteady gait and two episodes of vomiting. Three weeks ago, she had an upper respiratory tract infection that did not require treatment. There is no history of trauma or evidence of toxic ingestion. She takes no medications. Her temperature is 36.7°C (98°F), pulse is 120/min, respirations are 24/min, and blood pressure is 95/60 mm Hg. Examination shows slight horizontal nystagmus. Funduscopic examination shows no abnormalities. The patient is able to sit without support but is unable to walk without support. Deep tendon reflexes are normal, and sensation is intact. Which of the following is the most appropriate next step in management?

- ☐ A) Re-examination in 24 hours
- ☐ B) Complete blood count
- ☐ C) CT scan of the head
- ☐ D) EEG
- ☐ E) Lumbar puncture



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

17. A 37-year-old woman with a 10-year history of HIV infection comes to the physician because of headache and fever for 2 days. Her temperature is 38.9°C (102°F). Examination shows confusion and mild weakness of the left upper and lower extremities. Deep tendon reflexes are increased, and Babinski sign is present on the left. Her CD4+ T-lymphocyte count is 4/mm³ (Normal ≥500), and plasma HIV viral load is 60,000 copies/mL. Which of the following is the most likely finding on MRI of the brain?
- ☐ A) Diffuse leukoaraiosis with periventricular white matter changes
 - ☐ B) Markedly enlarged sulci with generalized loss of brain matter
 - ☐ C) Markedly enlarged ventricles with minimal changes in sulci
 - ☐ D) Mass lesion within the substance of the right frontal cortex with shift to the left of midline structures
 - ☐ E) Opaque mass over convexity of left frontal cortex with compression of brain substance



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

18. An 87-year-old man comes to the physician because of headaches over the past 6 weeks; they fluctuate in severity during the day and occasionally awaken him from sleep. He describes the pain as an aching pressure that is worse in the front. His vision is unchanged. He takes aspirin for cervical osteoarthritis. His son died 3 months ago. He is lethargic and yawns during the interview. His temperature is 36.6°C (97.8°F), pulse is 88/min and regular, and blood pressure is 140/90 mm Hg. Funduscopic examination shows cataracts. There is moderate stiffness of the neck with flexion and when the head is turned to the left and right. There is a pronator drift of the right upper extremity. Reflexes are 3+ at the biceps and triceps muscles and knees and 1+ at the ankles. Babinski sign is present bilaterally. He has a slow unsteady gait with short steps. No other abnormalities are noted. Which of the following is the most likely diagnosis?

- ☐ A) Craniopharyngioma
- ☐ B) Hypertensive encephalopathy
- ☐ C) Major depressive disorder
- ☐ D) Migraine
- ☐ E) Subdural hematoma
- ☐ F) Temporal arteritis



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

19. A 67-year-old woman is brought to the physician because of progressive imbalance and memory difficulties over the past year. Two years ago, she began to have episodes of sudden imbalance with a tendency to fall backwards. She has developed progressive slowness of thought and movement and has had brief episodes of double vision. Examination shows mildly decreased short-term memory and slowness of verbal responses. Downward gaze and upward gaze are markedly impaired. There is marked rigidity of the neck and mild rigidity of the limbs. The gait is broad based with short shuffling steps, and she easily loses her balance, falling backwards. All movements are performed slowly. A CT scan of the head without contrast shows moderate, diffuse atrophy. Which of the following is the most likely diagnosis?

- ☐ A) Creutzfeldt-Jakob disease
- ☐ B) Dementia, Alzheimer type
- ☐ C) Hydrocephalus
- ☐ D) Hypothyroidism
- ☐ E) Multi-infarct (vascular) dementia
- ☐ F) Parkinson disease with dementia
- ☐ G) Progressive supranuclear palsy
- ☐ H) Syphilis
- ☐ I) Wernicke-Korsakoff syndrome



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

20. An 80-year-old man briefly loses consciousness and falls to the floor in a grocery store. He awakens within 1 minute and feels dizzy, but after several minutes the dizziness passes. His pulse is 80/min; his blood pressure is 160/90 mm Hg while supine and 150/86 mm Hg while standing. Cardiac examination and ECG show no abnormalities. During the physical examination, when his head is turned to the right, he suddenly loses consciousness and motor tone. While unconscious, his pulse is 40/min. He awakens within 20 seconds and feels slightly dazed, but this feeling passes after 1 minute. Which of the following is the most likely diagnosis?

- ☐ A) Carotid sinus hypersensitivity
- ☐ B) Complex partial seizure disorder
- ☐ C) Idiopathic orthostatic hypotension
- ☐ D) Vasodepressor syncope
- ☐ E) Vertebrobasilar insufficiency



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

21. An 18-year-old woman is brought to the emergency department 30 minutes after a 5-minute generalized tonic-clonic seizure. She has a 1-year history of sudden jerking movements of her hands and arms; the episodes frequently occur while she is getting ready for work in the morning. On arrival, she is confused but gradually becomes oriented to person, place, and time. Her temperature is 36.9°C (98.4°F), pulse is 80/min, respirations are 20/min, and blood pressure is 114/68 mm Hg. On examination, ocular movements are full. The remainder of the examination shows no abnormalities. Deep tendon reflexes are 2+ throughout. In addition to valproic acid therapy, supplementation with which of the following is the most appropriate next step in management?

- ☐ A) Folic acid
- ☐ B) Riboflavin
- ☐ C) Vitamin B₁ (thiamine)
- ☐ D) Vitamin B₆ (pyridoxine)
- ☐ E) Vitamin D
- ☐ F) Vitamin E



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

22. A 67-year-old man with a 30-year history of alcoholism is brought to the emergency department 1 hour after he was found confused and lethargic by his daughter. He appears malnourished and ill. His temperature is 36°C (96.8°F), pulse is 96/min, and blood pressure is 150/80 mm Hg. Scattered rhonchi are heard. Abdominal examination shows no tenderness, a liver span of 18 cm, and a palpable spleen tip. Neurologic examination shows a disconjugate gaze. Which of the following is the most appropriate next step in management?

- ☐ A) Intensive physical therapy
- ☐ B) Intravenous glucose therapy
- ☐ C) Intravenous vitamin B₁ (thiamine) therapy
- ☐ D) Intravenous vitamin B₁₂ (cyanocobalamin) therapy
- ☐ E) Oral lactulose therapy



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

23. A 62-year-old woman is brought to the emergency department after an episode of flaccid paralysis and numbness of the left upper and lower extremities that lasted 2 hours. During the episode, she was unable to see objects in her left visual field. Neurologic examination now shows no abnormalities. A CT scan of the head shows no lesions. Which of the following is the most likely diagnosis?

- ☐ A) Arteriovenous malformation
- ☐ B) Cerebellar glioma
- ☐ C) Cerebral embolus
- ☐ D) Frontal lobe meningioma
- ☐ E) Lacunar infarct
- ☐ F) Migraine
- ☐ G) Rupture of a cerebral artery aneurysm
- ☐ H) Temporal lobe astrocytoma



24. A 25-year-old farm worker is brought to the emergency department because of the sudden onset of headache, dizziness, nausea, and vomiting while picking peaches. His pulse is 52/min, and blood pressure is 90/60 mm Hg. He has generalized weakness, diaphoresis, and pinpoint pupils. Which of the following precautions should medical professionals take while caring for this patient?

- ☐ A) Wearing protective goggles
- ☐ B) Wearing rubber gloves
- ☐ C) Wearing a surgical mask
- ☐ D) No precautions necessary



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

25. A 62-year-old man with amyotrophic lateral sclerosis comes to the physician because of a 1-month history of progressive shortness of breath. The shortness of breath is exacerbated when he is supine. His only medication is an 81-mg aspirin tablet daily. He has smoked two packs of cigarettes daily for 30 years but stopped 10 years ago. His pulse is 76/min and regular, respirations are 18/min while sitting and 26/min while supine, and blood pressure is 112/78 mm Hg. There is no jugular venous distention. Scattered rhonchi are heard on auscultation; they resolve with coughing. An S₄ is heard. Abdominal examination shows retractions of the abdominal walls with inspiration. There is no peripheral edema. Which of the following is the most likely cause of his shortness of breath?

- ☐ A) Bronchospasm
- ☐ B) Diaphragmatic paralysis
- ☐ C) Failure of intercostal muscle
- ☐ D) Left ventricular failure
- ☐ E) Panacinar emphysema
- ☐ F) Pleural effusions



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

26. A 14-month-old infant is brought to the physician because of four episodes of loss of consciousness over the past month. While in the office, the infant has a typical episode. Prior to onset, the infant became fretful when his pacifier was taken away. He cried for 30 seconds, then stopped crying and breathing and became rigid and cyanotic. There are no clonic movements and no response to noxious stimuli. His eyes deviate upward conjugately; the pupils do not respond to direct or consensual light. Faint heart sounds are heard on apical auscultation; a pulse cannot be located. He recovers without intervention and returns to normal behavior within 10 minutes. Which of the following is the most likely diagnosis?

- ☐ A) Absence seizure
- ☐ B) Adverse effect of medication
- ☐ C) Breath-holding episode
- ☐ D) Carotid artery trauma
- ☐ E) Narcolepsy-cataplexy
- ☐ F) Ventricular tachyarrhythmia



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

27. A 22-year-old woman comes to the physician because she has had a black spot that occludes her vision in her right eye for 1 day. She also has pain in the right eye. There is no history of trauma. Examination shows visual acuity of 20/20 in the left eye; the best corrected vision in the right eye is 20/200. The left pupil is normal; the right reacts sluggishly to direct light but consensually. Fundusoscopic examination shows no abnormalities. The eyes are not injected. Which of the following is the most likely site of the lesion?

- ☐ A) Cerebellum
- ☐ B) Cranial nerve II
- ☐ C) Cranial nerve III
- ☐ D) Cranial nerve IV
- ☐ E) Cranial nerve VI
- ☐ F) Extraocular muscles
- ☐ G) Medial longitudinal fasciculus
- ☐ H) Neuromuscular junction
- ☐ I) Retina



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

28. A 22-year-old college student comes to the student health center after an episode during which he found himself unable to move for several seconds after awakening. The episode occurred during the college's spring break 2 weeks ago, when he awoke after having fallen asleep on the beach. He has no history of similar symptoms. The patient is not having academic difficulties, although he admits to recent stress caused by the examinations that he took just before the spring break. His medical history is noncontributory. Vital signs are within normal limits. Examination shows no abnormalities. Which of the following is the most appropriate next step in management?

- ☐ A) Reassurance
- ☐ B) Polysomnography and multiple sleep latency test
- ☐ C) Psychiatric assessment
- ☐ D) Toxicology screening
- ☐ E) Sleep-deprived EEG
- ☐ F) MRI of the brain



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

29. An 82-year-old woman is brought to the physician because of a 6-month history of neck pain radiating to her left hand. She has also had neck stiffness and difficulty maintaining her balance while walking. She has osteoarthritis, Crohn disease, gastroesophageal reflux disease, and hyperlipidemia. Current medications include ranitidine, cholestyramine, mesalamine, and ibuprofen. Examination shows moderately decreased range of motion of the neck. There is mild weakness and atrophy of the hands and forearms and increased muscle tone of the lower extremities. The triceps reflexes are absent, the biceps and brachioradialis reflexes are intact, and the knee and ankle jerk reflexes are increased. Babinski sign is present bilaterally. Sensation to pinprick and temperature is decreased over the dorsal forearms and hands. Her gait is broad based and stiff. Which of the following is the most likely diagnosis?

- ☐ A) Bilateral brachial plexopathy
- ☐ B) Bilateral median and ulnar mononeuropathy
- ☐ C) Cervical spondylotic myelopathy
- ☐ D) Parasagittal meningioma
- ☐ E) Subacute combined degeneration



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

30. A 62-year-old man is brought to the emergency department because of a 1-week history of urinary hesitancy, leg weakness, and difficulty walking; he says that he falls occasionally. He has a 1-month history of sharp right-sided abdominal pain that is exacerbated by coughing, sneezing, and lying flat; the pain now radiates posteriorly to the interscapular region. At his last examination 3 days ago, testing for gallbladder disease was negative. He has hypertension treated with atenolol. His temperature is 37.1°C (98.8°F), pulse is 80/min, and blood pressure is 140/70 mm Hg. Examination shows tenderness to percussion over the midthoracic spine. Muscle strength is 4/5 in the iliopsoas, hamstrings, and anterior tibialis muscles bilaterally; it is otherwise full. Deep tendon reflexes are 2+ in the upper extremities and 3+ in the knees and ankles. Babinski sign is present bilaterally. Sensation to pinprick and temperature is decreased below the costal margin, and sensation to vibration is absent below the ankles. Intravenous administration of which of the following is the most appropriate next step in management?

- ☐ A) Ceftriaxone
- ☐ B) Dexamethasone
- ☐ C) Furosemide
- ☐ D) Heparin
- ☐ E) Mannitol



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

31. A 67-year-old woman is brought to the emergency department 1 hour after the sudden onset of right facial weakness. She has hypertension treated with lisinopril and hyperlipidemia treated with atorvastatin. She has smoked one pack of cigarettes daily for 35 years. Her pulse is 76/min, and blood pressure is 168/90 mm Hg. Examination shows right facial weakness of the forehead and lower face. There is decreased hearing and tinnitus on the right. Finger-nose testing shows mild ataxia on the right. An MRI of the brain is most likely to show an infarction in which of the following areas?

- ☐ A) Internal capsule
- ☐ B) Medulla
- ☐ C) Midbrain
- ☐ D) Pons
- ☐ E) Temporal lobe



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

32. A 52-year-old man is brought to the emergency department from work 30 minutes after the sudden onset of blindness. He is unable to remember all the events leading up to his visual loss, but he states that earlier in the day he saw his boss in a compromising position; he is uncomfortable speaking about this and quickly changes the subject. He has not had any eye pain but says that he has a headache. He has a history of migraines treated with sumatriptan. Ophthalmic examination shows no evidence of trauma, erythema, or infection. He flinches when the physician rapidly raises her hand toward his face. Funduscopic examination shows no evidence of papilledema, optic neuritis, or retinal hemorrhage or detachment. A CT scan of the head shows no abnormalities. On mental status examination, he appears unaffected by his vision loss. His speech is normal, and he does not have any delusional behavior. When his wife arrives at the emergency department, he asks if he can go home and inquires about when he can return to work. Which of the following is the most likely diagnosis?

- ☐ A) Acute stress disorder
- ☐ B) Brief psychotic disorder
- ☐ C) Conversion disorder
- ☐ D) Malingering
- ☐ E) Migraine
- ☐ F) Occipital lobe infarct



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

33. A 72-year-old woman is brought to the physician because of a 24-hour history of severe right ear pain, drooping of the right side of her face, and tearing of the right eye. She had a mild upper respiratory tract infection 1 week ago. She has type 2 diabetes mellitus currently treated with glyburide. She is in distress. Her temperature is 37°C (98.6°F). Examination shows rotatory nystagmus on lateral gaze. There is severe palsy of the right peripheral facial nerve with an inability to close the right eye. Examination shows a rash in the external auditory canal and hearing loss on the right. There is no nuchal rigidity. Which of the following is the most likely cause of these findings?

- ☐ A) *Borrelia burgdorferi* infection
- ☐ B) Herpes zoster
- ☐ C) Measles
- ☐ D) Mucormycosis
- ☐ E) *Pseudomonas aeruginosa* infection
- ☐ F) Staphylococcal infection



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

34. A 47-year-old woman is brought to the emergency department 1 hour after the onset of a severe headache that began suddenly while she was exercising at the gym. She rates her pain as a 10 on a 10-point scale and says it is present over her entire head. She has type 2 diabetes mellitus controlled with metformin. She is 172 cm (5 ft 7 in) tall and weighs 72 kg (158 lb); BMI is 25 kg/m². She is alert and oriented. Vital signs are within normal limits. On examination, the right pupil is 5 mm and not reactive to light. Meningismus is present. The remainder of the examination, including neurologic examination, shows no abnormalities. The most likely cause of these findings is a lesion in which of the following arteries?

- ☐ A) Anterior cerebral
- ☐ B) Basilar
- ☐ C) Middle cerebral
- ☐ D) Posterior communicating
- ☐ E) Proximal internal carotid



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

35. A 37-year-old man comes to the physician because of a 6-month history of difficulty walking and a tremor of the right arm. He says that his movements have been slower. He has hypertension, schizophrenia, and gastroesophageal reflux disease. Current medications include hydrochlorothiazide, fluphenazine, benztropine, and ranitidine. His blood pressure is 108/72 mm Hg. Examination shows increased muscle tone in the right upper extremity; there is a resting tremor of the right hand and decreased right arm swing. His gait is short based. Mental status examination shows a restricted affect. Which of the following is most likely to improve this patient's symptoms?

- ☐ A) Adding donepezil to the medication regimen
- ☐ B) Adding risperidone to the medication regimen
- ☐ C) Discontinuing benztropine
- ☐ D) Discontinuing fluphenazine
- ☐ E) Discontinuing ranitidine



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

36. A 25-year-old woman comes to the physician because of a 1-year history of depressed mood and episodes of daytime sleepiness despite getting an adequate amount of sleep at night. She takes daily naps that are marked by visions of bloody bodies before she falls asleep or upon awakening. She is frightened by these visions but cannot control her urge to sleep; she wakes up refreshed. She has smoked one pack of cigarettes daily for 8 years and drinks one to two cups of coffee daily. She does not use illicit drugs. She is 180 cm (5 ft 11 in) tall and weighs 75 kg (165 lb); BMI is 23 kg/m². Her temperature is 36°C (96.8°F), pulse is 80/min, and blood pressure is 120/80 mm Hg. Neurologic examination shows no focal findings. Mental status examination shows anxiety and a mildly depressed mood. An ECG shows no abnormalities. Which of the following is the most likely diagnosis?

- ☐ A) Brief psychotic disorder
- ☐ B) Major depressive disorder
- ☐ C) Narcolepsy
- ☐ D) Nightmare disorder
- ☐ E) Sleep apnea



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

37. A 56-year-old woman is brought to the emergency department 20 minutes after a simple partial seizure. A CT scan of the head shows a 1.5-cm, uniformly and intensely enhancing parasagittal mass in the left frontal region. Further evaluation for a primary lesion is negative. Examination shows mild weakness of the right lower extremity. Which of the following is the most likely diagnosis?

- ☐ A) Astrocytoma
- ☐ B) Ependymoma
- ☐ C) Glioblastoma multiforme
- ☐ D) Meningioma
- ☐ E) Schwannoma



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

38. A 72-year-old man with a history of atrial fibrillation comes to the emergency department 2 hours after the sudden onset of visual problems. He states that he was fine until this afternoon when he started walking into objects that were located on his right side. While driving himself to the hospital, he sideswiped a parked car. His only medication is daily aspirin. On arrival, his pulse is 68/min and irregular, and blood pressure is 140/70 mm Hg. Neurologic examination shows a right homonymous hemianopia. Pupils are equal and reactive, and fundoscopic examination shows no abnormalities. Muscle strength and sensation are normal. Mental status examination shows no abnormalities. Which of the following arteries is most likely to be occluded?

- | | |
|--|---|
| ☐ A) Basilar | ☐ G) Right anterior cerebral |
| ☐ B) Left anterior cerebral | ☐ H) Right internal carotid |
| ☐ C) Left internal carotid | ☐ I) Right middle cerebral |
| ☐ D) Left middle cerebral | ☐ J) Right ophthalmic |
| ☐ E) Left ophthalmic | ☐ K) Right posterior cerebral |
| ☐ F) Left posterior cerebral | |



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

39. A 35-year-old woman comes to the physician because of a 14-day history of progressively severe headaches, mild neck soreness, and photophobia. Twelve weeks ago, she was treated with x-ray radiation therapy and methotrexate, vincristine, procarbazine, and prednisone for stage IVB Hodgkin disease with a large anterior mediastinal mass; the only complication of therapy was a brief episode of fever and neutropenia. Her temperature now is 37.8°C (100°F). She does not appear acutely ill. Examination shows a mildly rigid neck. Neurologic examination shows no papilledema. Lymph nodes, liver, and spleen are not palpable. Cardiopulmonary examination and x-rays of the chest show normal findings. Serum glucose concentration is 70 mg/dL. Results of lumbar puncture show:

Glucose	15 mg/dL
Protein	120 mg/dL
Leukocyte count	500/mm ³
Segmented neutrophils	10%
Monocytes	90%
Erythrocyte count	0/mm ³

Which of the following is the most appropriate pharmacotherapy?

- ☐ A) Intramuscular meperidine
- ☐ B) Intravenous amphotericin B
- ☐ C) Intravenous ceftriaxone
- ☐ D) Intravenous methylprednisolone
- ☐ E) Intravenous penicillin G



40. A 52-year-old woman comes to the physician because of a 2-week history of moderate pain, numbness, and tingling in her right hand. She states that the pain is worse at night and is not relieved with acetaminophen or ibuprofen. She has a 10-year history of chronic lymphocytic (Hashimoto) thyroiditis treated with thyroid hormone therapy. Vital signs are within normal limits. On examination, tapping the palmar surface of the right wrist elicits a tingling sensation in the thumb and index fingers. Her serum thyroid-stimulating hormone concentration is 2 μ U/mL. Which of the following is the most likely mechanism of this patient's symptoms?

- ☐ A) Immunologic damage to the brachial nerve
- ☐ B) Immunologic damage to the median nerve
- ☐ C) Immunologic damage to the radial nerve
- ☐ D) Immunologic damage to the ulnar nerve
- ☐ E) Mechanical entrapment of the brachial nerve
- ☐ F) Mechanical entrapment of the median nerve
- ☐ G) Mechanical entrapment of the radial nerve
- ☐ H) Mechanical entrapment of the ulnar nerve



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

41. After falling from a horse during a steeplechase, a 42-year-old man has immediate paralysis in all four extremities and has loss of sensation below the clavicles bilaterally. On examination by paramedics, he is alert, dyspneic, and unable to move his extremities or trunk. His pulse is 98/min, respirations are 28/min, and blood pressure is 150/90 mm Hg. After transport to a trauma center, his sensory examination returns to normal, and he regains normal strength of the arms and forearms, although the hands and lower extremities remain weak. Examination shows bladder distention. An MRI of the cervical spine shows contusion with edema of the spinal cord at C3 and C4. Plain x-rays of the cervical spine show a nondisplaced fracture of the vertebral body of C3. Which of the following is the most appropriate next step in management?

- ☐ A) Intravenous ceftriaxone therapy
- ☐ B) Intravenous labetalol therapy
- ☐ C) Intravenous mannitol therapy
- ☐ D) Intravenous methylprednisolone therapy
- ☐ E) Oral prednisone therapy
- ☐ F) Surgical exploration of the cervical spine



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

42. A 47-year-old man is brought to the emergency department by police after he was found in a state of confusion outside a convenience store. He says he feels anxious. He has not had fever, headache, abdominal pain, nausea, or vomiting. He has a 3-year history of uncontrolled hypertension. He has no allergies and takes no medications. He does not smoke. He usually drinks 12 to 14 beers daily. He says he has never used illicit drugs. He is disheveled and agitated and picks at imaginary things in the air. He is disoriented to person, place, and time. His temperature is 37.5°C (99.5°F), pulse is 110/min, and blood pressure is 160/105 mm Hg. Examination shows diaphoresis and tremulousness. There is seborrhea and rosacea of the face. There are no signs of head trauma. The neck is supple. There is diffuse hyperreflexia. Babinski sign is absent bilaterally. Which of the following is the most likely diagnosis?

- ☐ A) Acute subdural hematoma
- ☐ B) Alcohol intoxication
- ☐ C) Brief psychotic disorder
- ☐ D) Cocaine intoxication
- ☐ E) Delirium tremens
- ☐ F) Encephalitis
- ☐ G) Hypoglycemia
- ☐ H) Meningitis



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

43. A 6-month-old boy is admitted to the hospital for evaluation of spasms. The child's father and a paternal aunt are mildly retarded and have seizures. The father has acneiform lesions on his cheeks, but without comedones. Examination of the patient shows five 1- to 2-cm depigmented areas on the chest and flexor surfaces of the upper and lower extremities. A CT scan of the head shows a subependymal calcification near the foramen of Monro on the left. Which of the following is the most likely diagnosis?
- ☐ A) Idiopathic hypercalcemia
 - ☐ B) Neurofibromatosis
 - ☐ C) Sturge-Weber syndrome
 - ☐ D) TORCH syndrome
 - ☐ E) Tuberous sclerosis



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

44. A 62-year-old man comes to the physician because of a 4-month history of progressive difficulty climbing stairs, generalized muscle pain, and dry mouth. He has hypertension, type 2 diabetes mellitus, and hyperlipidemia. Current medications include atorvastatin, atenolol, and glyburide. He has smoked two packs of cigarettes daily for 30 years. At his last visit 6 weeks ago, an x-ray of the chest showed a solitary lung nodule. Today, his pulse is 80/min, respirations are 14/min, and blood pressure is 136/82 mm Hg. Examination of the lower extremities shows bilateral proximal weakness and hyporeflexia. Electromyography and nerve conduction studies show an incremental increase in the amplitude of evoked motor response. Which of the following is most likely to have prevented these symptoms?

- ☐ A) Smoking cessation
- ☐ B) Daily aspirin therapy
- ☐ C) Discontinuing atenolol therapy
- ☐ D) Discontinuing atorvastatin therapy
- ☐ E) More aggressive treatment of high blood pressure



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

45. A 30-year-old construction worker comes to the physician because of constant midlower back pain for 1 month. He has no leg pain. Straight-leg raising causes discomfort only in the low back. Sensorimotor examination is normal. Which of the following is the most appropriate next step in management?

- ☐ A) Restriction of heavy lifting until the pain resolves and instruction in proper body mechanics
- ☐ B) Bed rest and oral analgesics
- ☐ C) Vocational rehabilitation and change in occupation
- ☐ D) Epidural injection of corticosteroids and nonsteroidal anti-inflammatory drugs
- ☐ E) Intervertebral disc excision and fusion



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

46. A 32-year-old man comes to the physician because of a 6-month history of deep aching neck and shoulder pain, progressive weakness of his hands, and poor coordination. Two years ago, he had a whiplash injury that resolved without treatment over a 3-month period. Neurologic examination shows hyporeflexia of the upper extremities. He has atrophy of intrinsic hand muscles bilaterally. There is asymmetric decreased strength of grip and finger abduction and extension. Sensation to pinprick is decreased over the upper torso and shoulders. Which of the following is the most likely cause of these findings?

- ☐ A) Amyotrophic lateral sclerosis
- ☐ B) Anterior spinal artery syndrome
- ☐ C) Brown-Séquard syndrome
- ☐ D) Posterior cord syndrome
- ☐ E) Subacute combined degeneration
- ☐ F) Syringomyelia



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



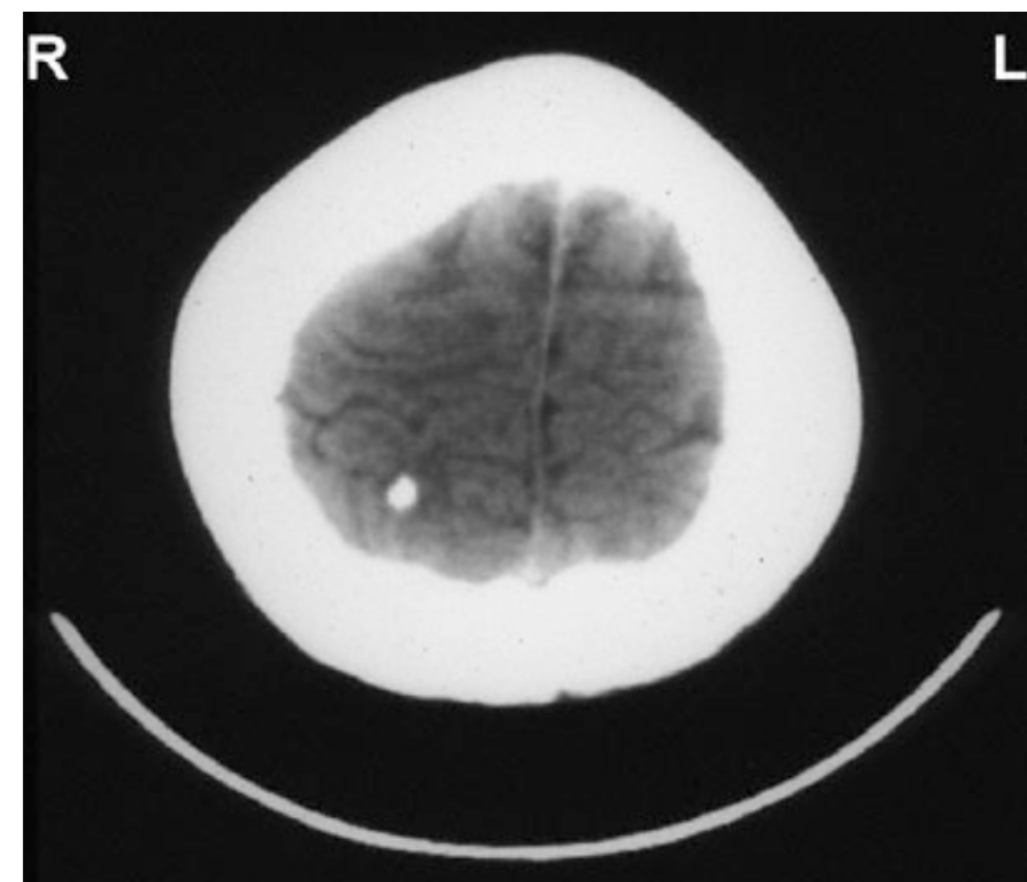
Help



Pause

47. A previously healthy 13-year-old girl is brought to the emergency department 1 hour after a generalized tonic-clonic seizure that occurred at school. Her teacher noted that the seizure was preceded by jerking of the right arm. She had a similar episode 8 months ago. Examination at that time showed no abnormalities, and no treatment was begun. She was born in Mexico, and she and her parents have made occasional trips to rural Mexico; their last visit was 2 years ago. She was lethargic on arrival to the emergency department, but she is now awake and alert. Her temperature is 37°C (98.6°F). The neck is supple. Neurologic examination shows no abnormalities. Her leukocyte count is 6400/mm³. A CT scan of the head is shown. Which of the following is the most appropriate screening test for family members of this patient?

- ☐ A) Examination of the stool for ova and parasites
- ☐ B) Rapid plasma reagin
- ☐ C) Stool culture for *Coccidioides immitis*
- ☐ D) Stool culture for *Histoplasma capsulatum*
- ☐ E) Stool culture for *Toxoplasma gondii*
- ☐ F) Urine culture for cytomegalovirus
- ☐ G) CT scan of the head



48. A 6-year-old girl is brought to the physician because of a 1-week history of progressive weakness and numbness of the legs. This morning, she had urinary incontinence. One year ago, she underwent resection of a medulloblastoma and placement of a ventriculoperitoneal shunt for residual hydrocephalus; her postoperative course had been uncomplicated. Currently, she is drowsy and irritable. Her temperature is 37.5°C (99.5°F), pulse is 82/min, respirations are 24/min, and blood pressure is 126/54 mm Hg. Funduscopy examination shows no abnormalities. Examination of the lower extremities shows weakness and diffuse sensory loss; deep tendon reflexes are decreased. Which of the following is the most likely underlying mechanism for these symptoms?

- ☐ A) Local recurrence of the medulloblastoma
- ☐ B) Metastatic spread of the cancer hematogenously
- ☐ C) Metastatic spread of the cancer through cerebrospinal fluid pathways
- ☐ D) Radiation necrosis of the brain stem
- ☐ E) Thrombosis of the anterior spinal artery
- ☐ F) Thrombosis of the superior sagittal sinus



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

49. A 45-year-old woman is brought to the emergency department 30 minutes after a generalized tonic-clonic seizure lasting 1 minute. She had a headache 6 hours ago and nausea and vomiting 2 hours ago. She has no history of seizures and has had no medical problems other than hypertension treated with enalapril and hydrochlorothiazide. Her temperature is 37.2°C (99°F), pulse is 100/min and regular, respirations are 16/min, and blood pressure is 230/140 mm Hg. She appears drowsy but is responsive to commands. Examination shows no abnormalities other than the funduscopic findings shown. Which of the following is the most appropriate initial pharmacotherapy?

- ☐ A) Clonidine
- ☐ B) Furosemide
- ☐ C) Lisinopril
- ☐ D) Nitroprusside
- ☐ E) Propranolol



50. A 42-year-old woman comes to the physician because of a 4-day history of progressive weakness and numbness in her legs and blurred vision and discomfort in both eyes. She has no history of serious illness and takes no medications. Visual acuity is 20/80 bilaterally. Examination shows bilateral optic disc edema. Muscle strength is 5/5 in the upper extremities and 4/5 in the lower extremities. Deep tendon reflexes are 2+ in the upper extremities and 4+ in the lower extremities. Babinski sign is present bilaterally. A lumbar puncture is performed. Analysis of the cerebrospinal fluid shows:

Glucose	80 mg/dL
Total protein	105 mg/dL
WBC	50/mm ³
RBC	5/mm ³
Oligoclonal bands	0 bands

An MRI of the brain shows multiple periventricular white matter lesions. An MRI of the thoracic spine shows an abnormal T2 signal and mild enhancement within the thoracic cord spanning T3 to T6. Which of the following abnormalities is most likely to be noted on further testing in this patient?

- ☐ A) Abnormal echocardiography
- ☐ B) Antinuclear antibodies in the serum
- ☐ C) Aquaporin-4 autoantibodies in the cerebrospinal fluid
- ☐ D) Decreased peripheral nerve conduction velocities
- ☐ E) Decreased serum copper concentration



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause